FinTech Research Report

Indian FinTech Ecosystem and Data Analytics – Case Study of Groww

1. Introduction

Financial Technology, or **FinTech**, refers to the use of technology to provide and improve financial services. In India, FinTech has transformed how people invest, trade securities, make payments, borrow money, and manage their finances.

This report provides an overview of the Indian FinTech ecosystem and examines **Groww** as a case study. Groww provides investment products including mutual funds, stocks, ETFs, IPOs, derivatives and other financial services. Its investor-relations information reported **22.7 million total transacting users and ₹3.56 trillion in customer assets as of August 1, 2026**, demonstrating the scale at which a digital investment platform operates.

Groww Investor Relations

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2. Indian FinTech Ecosystem

The Indian FinTech ecosystem consists of several interconnected participants:

- FinTech platforms and applications
- Banks
- Stock brokers
- Stock exchanges
- Clearing corporations
- Depositories
- Payment providers
- Insurance companies
- Asset management companies
- Regulatory authorities
- Investors and customers

Technology connects these participants and enables financial transactions to be processed quickly and digitally.

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3. Brokerage Platforms

A brokerage platform allows investors to buy and sell securities through recognised stock exchanges.

A typical digital brokerage platform provides:

- Trading accounts
- Market data
- Order placement
- Portfolio tracking
- Transaction reports
- Investment research
- Customer support

When an investor places an order through a platform such as Groww, the order is routed to the relevant market infrastructure for execution.

According to SEBI, an investor generally needs a **trading/broking account, demat account, and bank account** to participate in the securities market.

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4. Demat Accounts

A **Demat account** stores securities such as shares and other eligible securities in electronic form.

Instead of holding physical share certificates, investors' securities are maintained electronically through the depository system.

India has two major depositories:

1. **NSDL – National Securities Depository Limited**
2. **CDSL – Central Depository Services (India) Limited**

SEBI describes NSDL and CDSL as India's two depositories, while Depository Participants (DPs) act as intermediaries through which investors access depository services.

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5. NSDL and CDSL

Depositories are an important part of India's financial infrastructure.

Their major functions include:

- Holding securities electronically

- Transferring securities between accounts
- Supporting settlement
- Processing corporate actions
- Supporting pledges and related services
- Providing electronic records of securities ownership

SEBI explains that NSDL introduced dematerialisation in India, while CDSL provides electronic holding and settlement services.

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6. Role of SEBI

SEBI – Securities and Exchange Board of India is the primary regulator of India's securities market.

Its objectives include:

- Protecting investors
- Regulating securities markets
- Promoting market development
- Establishing rules for market intermediaries
- Improving transparency
- Reducing market-related risks

SEBI also establishes frameworks governing trading, clearing, settlement, depositories and intermediaries.

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7. Trading Lifecycle

A simplified trading lifecycle is:

Investor → Broker → Stock Exchange → Clearing Corporation → Depository → Investor's Demat Account

Step 1 – Order Placement

The investor places a buy or sell order through a trading platform.

Step 2 – Order Routing

The broker sends the order to the appropriate stock exchange.

Step 3 – Order Matching

The exchange matches compatible buy and sell orders based on price and other applicable rules.

Step 4 – Trade Confirmation

Once matched, the trade is executed and recorded.

Step 5 – Clearing

The clearing corporation calculates the obligations of the participants.

Step 6 – Settlement

Funds and securities are exchanged according to the applicable settlement cycle.

Step 7 – Demat Update

The securities are credited to or debited from the investor's demat account.

SEBI describes the process as involving the investor, broker, exchange, clearing corporation and depository, with NSDL/CDSL transferring securities as part of settlement.

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8. Settlement Process

India has significantly shortened its securities settlement cycle.

The regular equity market moved to **T+1 settlement**, meaning a trade executed on day T is normally settled on the next working day. SEBI notes that complete migration to T+1 settlement took place in January 2023. A T+0 beta settlement framework was subsequently introduced for eligible securities.

A simplified T+1 process is:

T – Trade → T+1 – Clearing & Pay-in/Pay-out → Securities/Funds Settled

Shorter settlement cycles can reduce settlement risk and allow investors to receive securities or funds more quickly.

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9. Market Participants

Major participants in the Indian securities market include:

Participant	Role

Participant	Role
Investors	Buy and sell financial securities
Stock Brokers	Provide trading access
Stock Exchanges	Match and execute orders
Clearing Corporations	Calculate and manage settlement obligations
NSDL/CDSL	Hold and transfer securities electronically
Depository Participants	Provide depository services to investors
Banks	Handle payments and funds
SEBI	Regulates the securities market
Asset Management Companies	Manage mutual funds
FinTech Platforms	Provide technology-driven financial services

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10. Case Study – Groww

Groww is a technology-driven Indian investment platform focused on simplifying investing for customers.

Its product ecosystem includes:

- Mutual funds
- SIPs
- Stocks
- ETFs
- IPOs
- Margin Trading Facility
- Equity derivatives
- Commodity derivatives
- API trading
- Credit products

Its investor-relations page shows that the platform operates at significant scale, reporting ₹45.52 billion in mutual-fund SIP inflows and ₹3.33 trillion in stocks turnover as of August 1, 2026.

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11. How Groww Uses Data Analytics

Data analytics is important for a digital FinTech platform because millions of customers generate large amounts of data through searches, investments, transactions, product interactions and customer-support activities.

A simplified analytics flow is:

Customer Activity → Data Collection → Data Processing → Analytics → Customer Insights → Business Action

Examples of potentially useful data include:

- Product searches
- Investment transactions
- SIP activity
- Portfolio activity
- Customer engagement
- Content interaction
- Customer-support interactions
- Product usage
- Customer retention

Groww publicly describes a customer-focused approach in which teams use real customer conversations and feedback to identify problems and improve products.

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12. Data Analytics for Customer Experience

One major use of analytics is improving the customer experience.

For example, customer behaviour can help a FinTech platform understand:

- Which products customers are interested in
- Which features are difficult to use
- What information customers search for
- Where customers drop out of a process
- Which support issues occur frequently

These insights can be used to improve the application interface, educational content, notifications and customer support.

Groww also reports using technology and AI initiatives to improve customer interactions and address customer research and support requirements.

The overall objective is:

Understand Customers → Identify Problems → Improve Product → Better Customer Experience

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13. Data Analytics for Business Decisions

Analytics can also support management decisions.

Important business metrics can include:

- Total transacting users
- Customer assets
- SIP inflows
- Trading turnover
- Product adoption
- Customer retention
- Revenue
- Profitability
- Customer-support performance

Groww publishes several of these operating metrics through its investor-relations platform, allowing stakeholders to monitor the scale and performance of its business.

Management can use such information to decide:

- Which products should be improved?
- Which customer segments are growing?
- Where should technology investment be increased?
- Which features have strong adoption?
- Where are customers experiencing problems?
- How can customer retention be improved?

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14. Example Analytics Use Case

Scenario

Suppose a large number of customers search for mutual funds but do not complete an investment.

Data

The company could analyse:

- Number of searches
- Fund pages viewed
- Time spent on pages
- SIP calculator usage
- Application drop-off points
- Customer-support queries

Analysis

A data analyst could identify where customers are leaving the investment journey.

For example:

Search → Fund Page → Investment Process → Drop-off

If a large percentage of customers leave at the investment stage, the company could investigate whether the process is confusing or requires too many steps.

Business Action

The product team could simplify the process, improve explanations, or redesign the relevant screen.

Expected Outcome

Lower drop-off → More completed investments → Better customer experience → Improved business performance

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15. Data Analytics and Risk Management

FinTech platforms also need strong controls because they process financial transactions and sensitive customer information.

Analytics can support:

- Transaction monitoring
- Fraud detection
- Unusual activity detection
- System monitoring
- Customer verification

- Risk analysis
- Operational monitoring

Security is particularly important because a digital investment platform handles financial and personal information.

Groww states on its public impact page that it uses industry-grade multi-factor authentication and reports zero data breaches over the preceding eight years as of June 30, 2026.

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16. Importance of Data Analysts in FinTech

Data Analysts play an important role in FinTech organisations.

Their responsibilities may include:

1. Collecting and cleaning financial data.
2. Analysing customer behaviour.
3. Creating business intelligence dashboards.
4. Monitoring KPIs.
5. Identifying trends and anomalies.
6. Supporting product decisions.
7. Measuring customer retention.
8. Analysing transaction behaviour.
9. Supporting risk and fraud analysis.
10. Communicating insights to business teams.

A Data Analyst therefore acts as a bridge between **raw data and business decisions**.

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17. Key FinTech Technologies

Modern FinTech platforms commonly use technologies such as:

- Web and mobile applications
- REST APIs
- Cloud computing
- Relational databases
- Data warehouses
- ETL/ELT pipelines

- Business intelligence tools
- Machine learning
- Artificial intelligence
- Authentication systems
- Monitoring and logging

These technologies work together to process financial transactions and convert operational data into useful information.

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18. Overall FinTech Data Flow

The complete high-level flow can be represented as:

Customer

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FinTech Application

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Authentication & API Layer

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Business Logic

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Trading / Investment Systems

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Banks / Exchanges / Clearing / Depositories

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Transaction & Customer Data

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Data Pipeline

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Data Warehouse / Database

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Analytics & BI

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Dashboards and Business Insights

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Improved Products and Customer Experience

This creates a continuous feedback loop where business teams use data to improve the platform.

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19. Key Findings

The research provides the following key findings:

- India's FinTech ecosystem connects customers, banks, brokers, exchanges, clearing corporations and depositories.
- Demat accounts enable electronic ownership of securities.
- NSDL and CDSL are India's two major securities depositories.
- SEBI regulates and develops India's securities market.
- The Indian equity market operates primarily on a T+1 settlement cycle, with T+0 available under a separate framework for eligible securities.
- Digital investment platforms generate large amounts of customer and transaction data.
- Data analytics can improve customer experience by identifying user behaviour and product problems.
- Analytics can support business decisions through KPIs such as users, assets, SIP inflows and trading activity.
- Data-driven risk monitoring and security are essential for FinTech platforms.

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20. Conclusion

The Indian FinTech ecosystem combines financial institutions, technology platforms and market infrastructure to provide faster and more accessible financial services.

Groww demonstrates how a technology-driven investment platform can operate at large scale while using customer and business data to understand users, monitor performance and improve products. Its published operating metrics show the importance of measuring users, customer assets, SIP inflows and trading activity.

For a Data Analyst, understanding the **business process behind the data** is just as important as knowing tools such as Python, SQL, Excel or Power BI.

The central idea of this research is:

FinTech Activity → Data → Analytics → Insights → Business Decisions → Better Customer Experience

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